

Week 5
Trajectories

ENN581 Robot Motion, Control and Planning

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School of Electrical Engineering and Robotics

ACKNOWLEDGEMENT OF TRADITIONAL OWNERS

QUT acknowledges the Turrbal and Yugara, as the First Nations owners of the lands where QUT now stands. We pay respect to their Elders, lores, customs and creation spirits. We recognise that these lands have always been places of teaching, research and learning.

QUT acknowledges the important role Aboriginal and Torres Strait Islander people play within the QUT community.

Most have done well

Check if an IK solution exists

Comfortable with ETS?

Comfortable with notation?

Assessment feedback

Check the limits on joint angles

How many of you used CoppeliaSim to verify?

- LM

Many used the Levemberg-Marquadt solver for IK. Do you know how it works?

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Last week, we worked on

- Parallel robots
  - Various constructions of parallel chain mechanisms
  - Comparison with serial chain mechanisms
- Mobility of parallel robots
- Workspace of parallel robots
- Forward kinematics of parallel robots
- Inverse kinematics of parallel robots

ANY QUESTIONS?

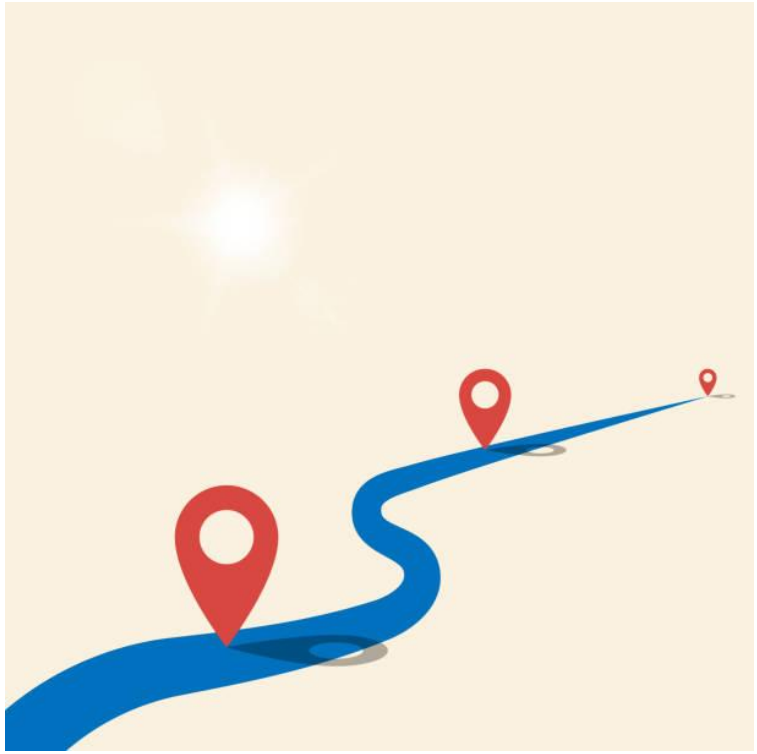
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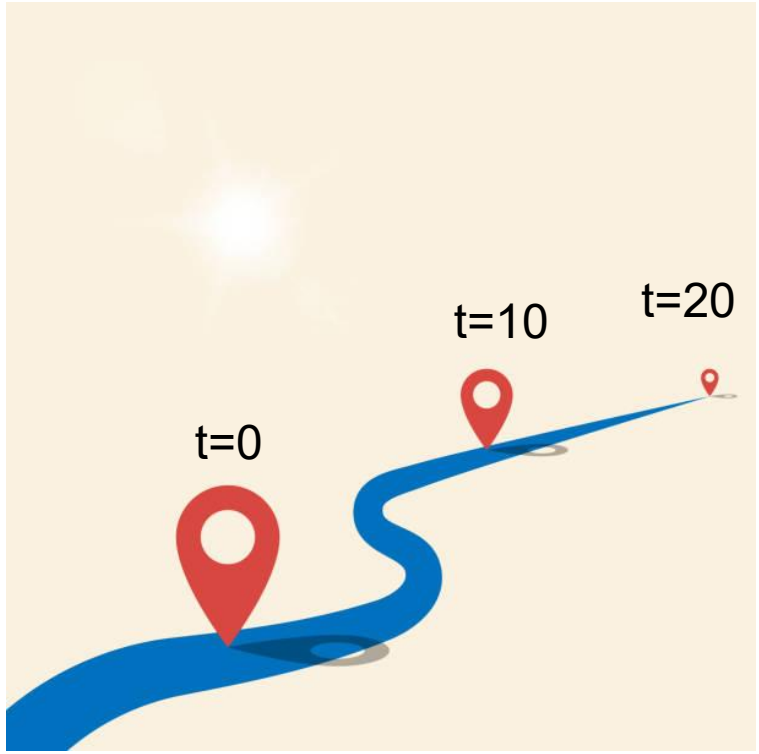
Today's lecture

- Trajectories
  - Configuration (Joint) space
  - Task (Cartesian) space
- Generating trajectories in configuration space
  - Polynomial functions
  - Linear functions
  - Blending
- Generating trajectories in task space
  - Straight-line and other interesting paths

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What is a trajectory?





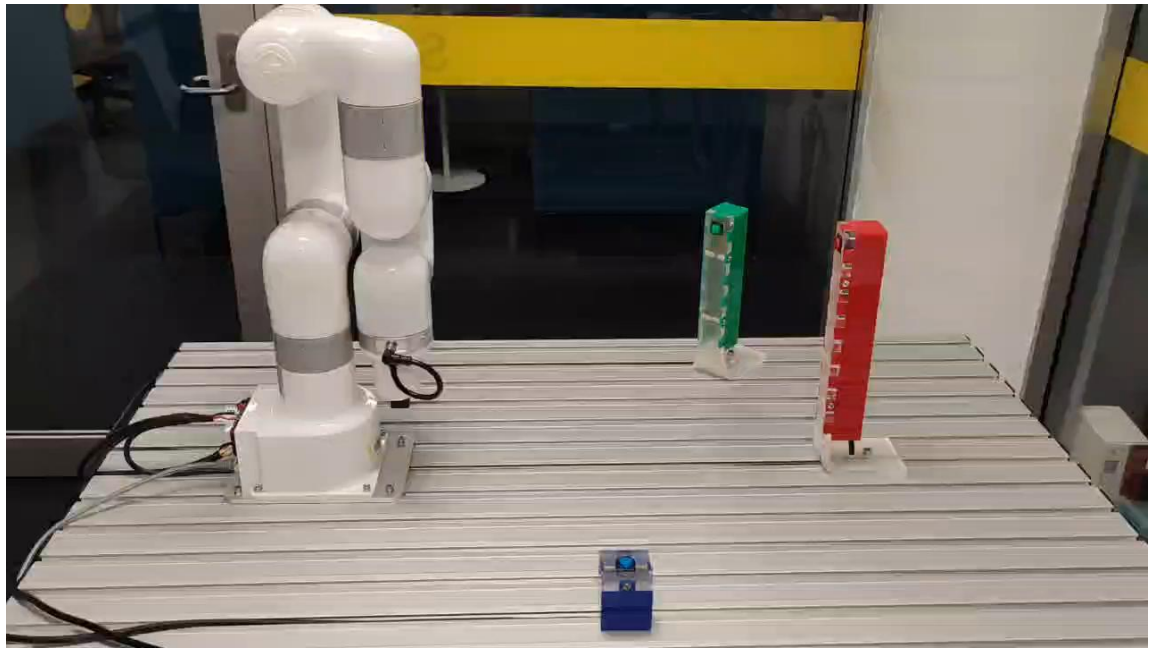
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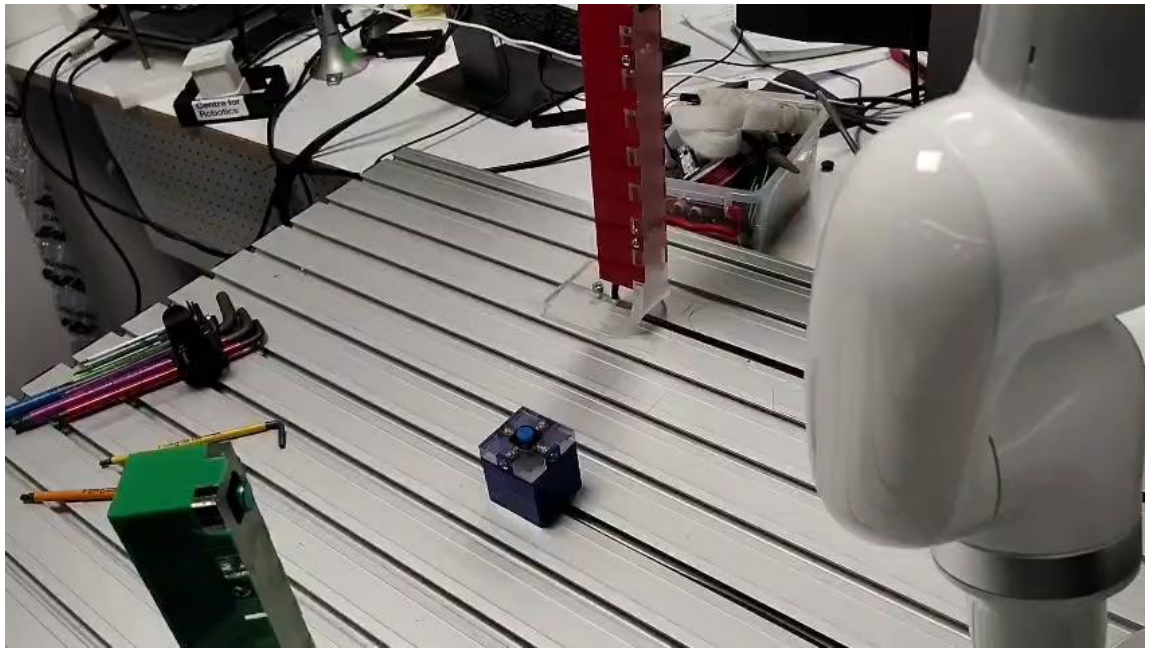
What is a trajectory?

A trajectory is a time sequence of position, (velocity and acceleration), for every degree of freedom.

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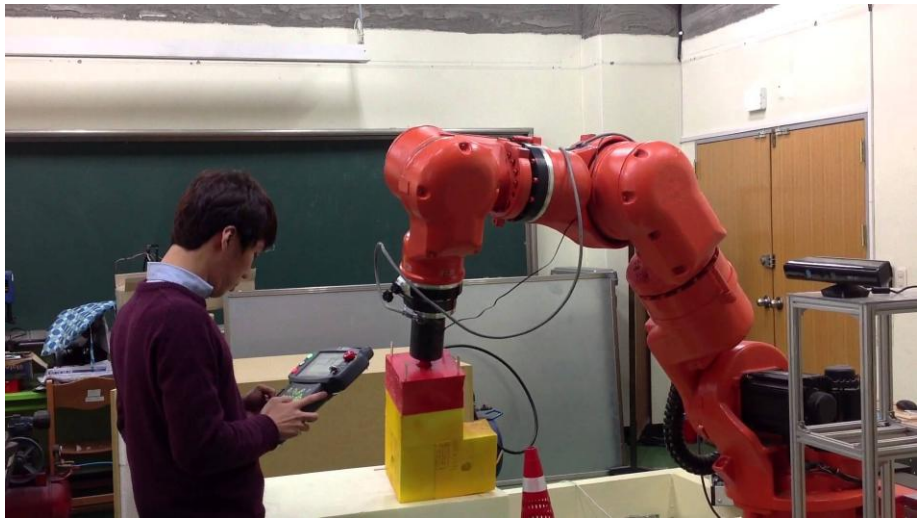
We want the tool to move from an initial pose to a final pose





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Teach mode operation



The operator jogs the robot through various positions manually (e.g., with a teach pendant or by holding the end effector). During this movement, positions of the joints are stored. These positions are recalled during operation.

No need to perform any kinematic calculations.

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What if the trajectory must be generated?

- Given: the initial and final pose of the tool.
- Aim: to figure out the motion of the tool between these points (frames).

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Configuration (Joint) Space Trajectories

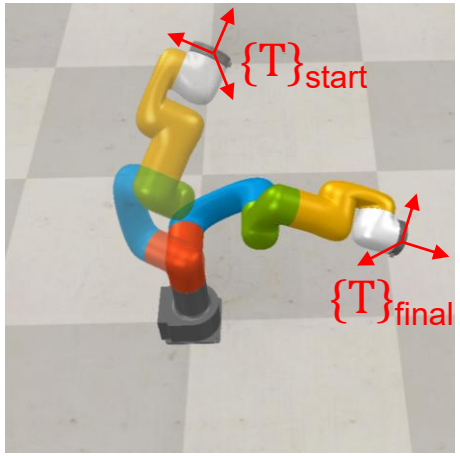
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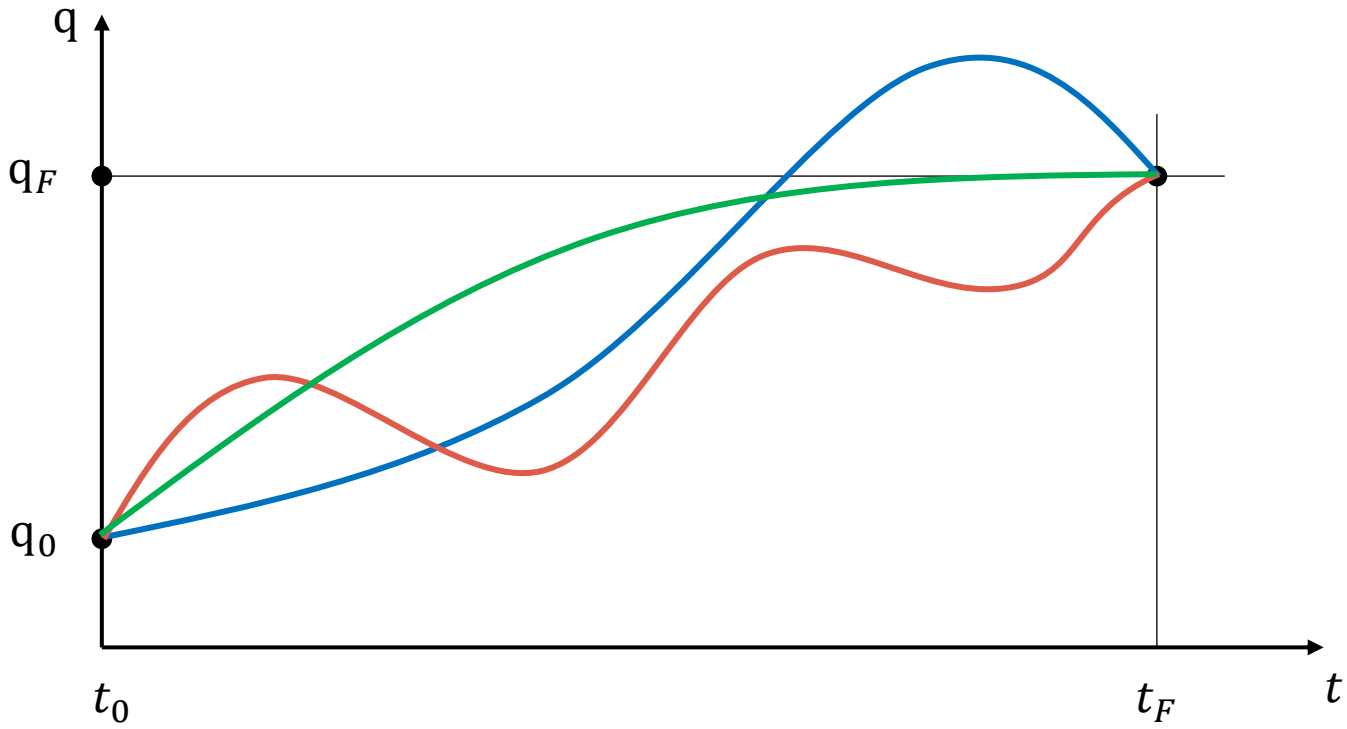
Approach 1: Think in terms of the joint coordinates

- Convert the initial and final points into joint positions using IK.
- Generate a trajectory in between, using joint positions.

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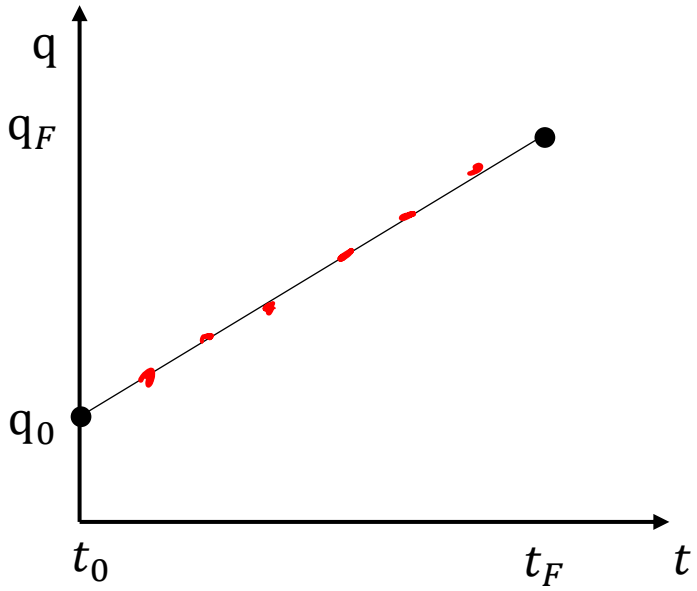
Which path should one choose?



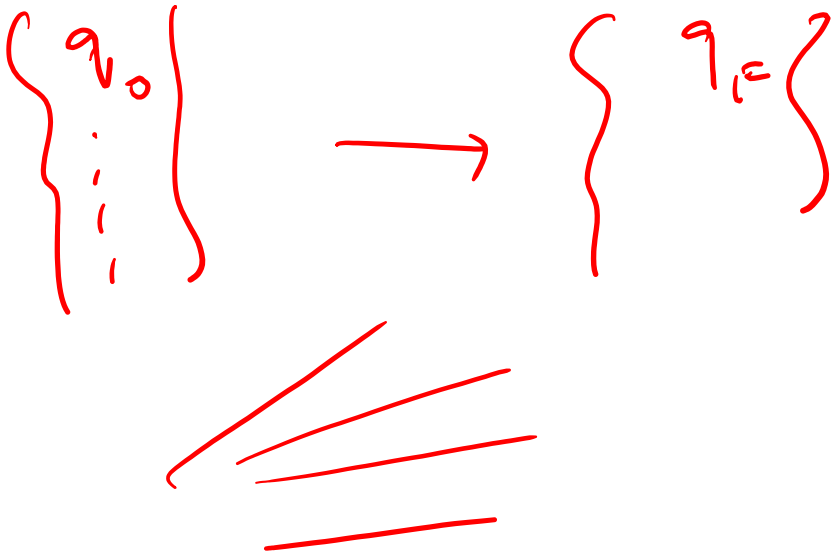


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Linear trajectories



- The simplest trajectory is a straight-line interpolation.



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Trajectories can also be thought of as polynomial functions

$$q(t) = \sum_{k=0}^{n} a_k t^k$$

cubic

$at^3 + bt^2 + ct + d$

q_0 q_f

Say, we are given the initial and final position. The manipulator starts at rest and must come to a stop at the final position.

q_f q_0

How many conditions on motion does the above statement specify?

What is the simplest function that satisfies these conditions?

A cubic polynomial

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Cubic polynomial function

Joint position q is described by a cubic polynomial of time t

$$\mathrm{q} = a_0 + a_1 t + a_2 t^2 + a_3 t^3$$

There are 4 unknowns. We need 4 equations to solve for them.
These are provided by initial and final conditions on position and velocity.

$$\mathrm{q}(t = 0) = \mathrm{q}_0 \qquad \mathrm{q}(t = t_F) = \mathrm{q}_F \qquad \dot{\mathrm{q}}(t = 0) = \dot{\mathrm{q}}_0 \qquad \dot{\mathrm{q}}(t = t_F) = \dot{\mathrm{q}}_F$$

The velocity (first derivative) is expressed as $\dot{\mathrm{q}} = a_1 + 2a_2 t + 3a_3 t^2$

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Example

Refer to this week's Jupyter notebook for how to solve this problem programmatically.

Consider a manipulator with a single revolute joint. The manipulator starts from rest at $\mathrm{q} = 10^{\circ}$. It must move to $\mathrm{q} = 50^{\circ}$ in 2 seconds. Determine the coefficients of a cubic that accomplishes this motion and brings the manipulator to a stop at the goal.

$q(0) = 10^{\circ}$

$q(2) = 50^{\circ}$

$\dot{q}(0) = 0$

$\dot{q}(2) = 0$

$q = a_0 + a_1 t + a_2 t^2 + a_3 t^3$

$q(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3$

$a_0 = ?$ $a_1 = ?$ $a_2 = ?$ $a_3 = ?$

$\downarrow 10^{\circ}$ $\downarrow 0$ 30 -10

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Example

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Consider a manipulator with a single revolute joint. The manipulator starts from rest at $\mathrm{q} = 10^{\circ}$. It must move to $\mathrm{q} = 50^{\circ}$ in 2 seconds. Determine the coefficients of a cubic that accomplishes this motion and brings the manipulator to a stop at the goal.

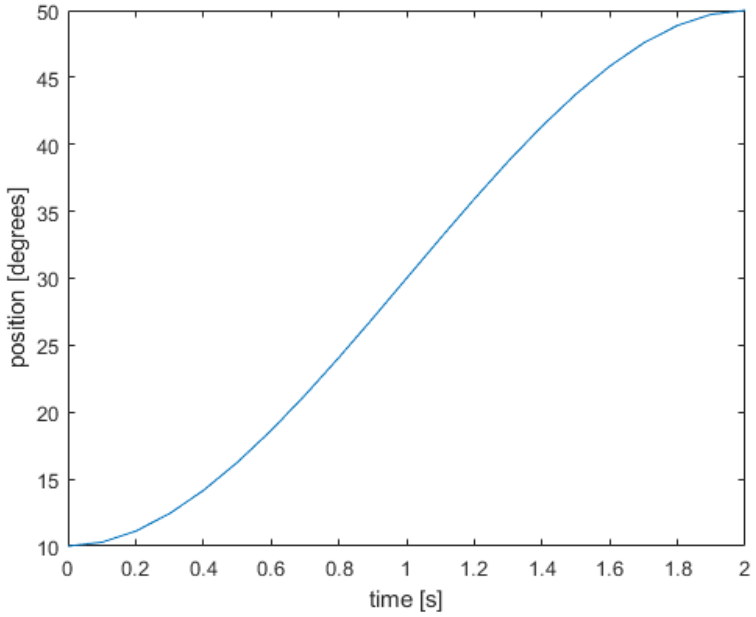
$$\mathrm{q} = a_0 + a_1 t + a_2 t^2 + a_3 t^3$$

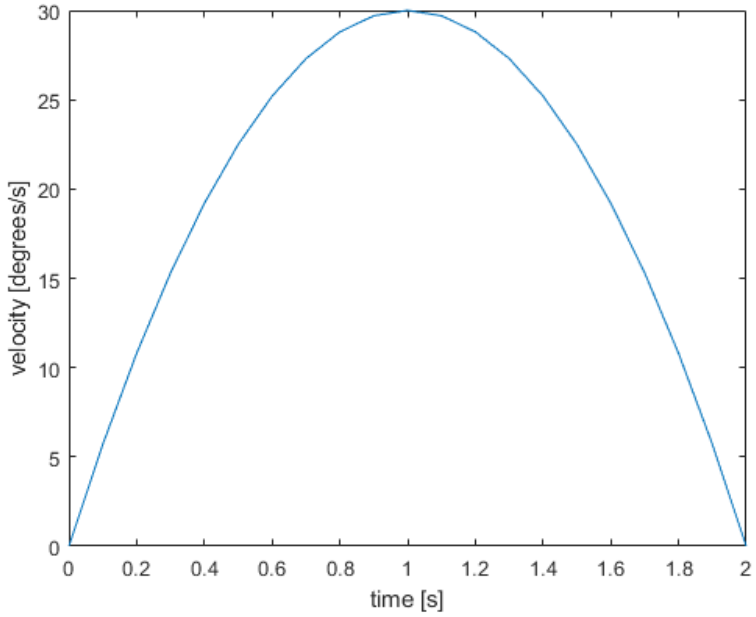
$$\mathrm{q}_0 = 10^{\circ} \qquad \mathrm{q}_F = 50^{\circ} \qquad \dot{\mathrm{q}}_0 = 0 \qquad \dot{\mathrm{q}}_F = 0 \qquad t_F = 2\ \mathrm{s}$$

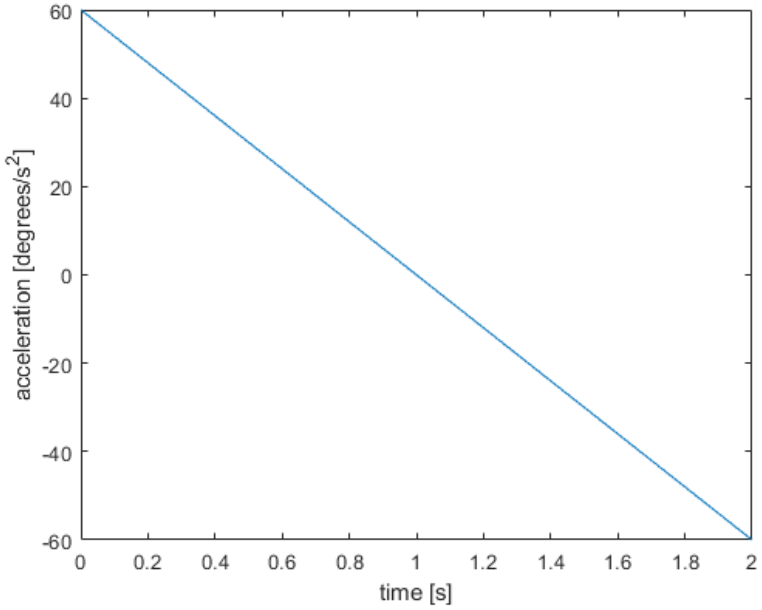
$$a_0 = 10^{\circ} \qquad a_1 = 0\ ^{\circ}/s \qquad a_2 = 30\ ^{\circ}/s^2 \qquad a_3 = -10\ ^{\circ}/s^3$$

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Position, velocity and acceleration







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What if the robot has more joints?

- Each joint has its own cubic polynomial.
- This is also referred to as multi-axis trajectory generation.

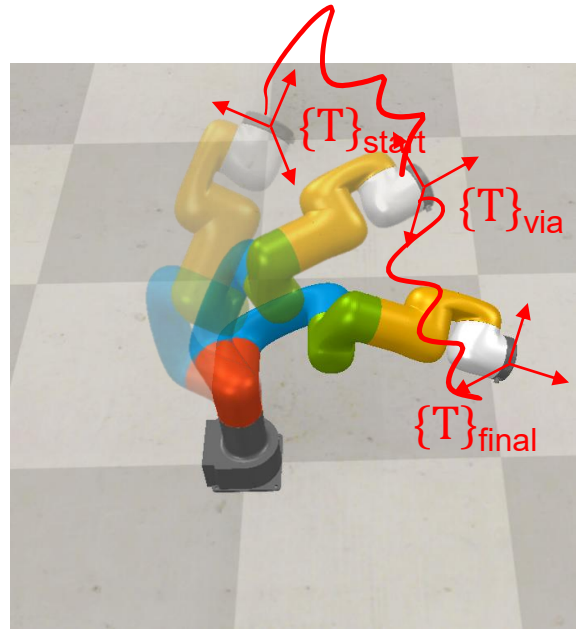
- Typically, a computer program sets a path-update rate, based on which, intermediate positions are determined.

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Via points

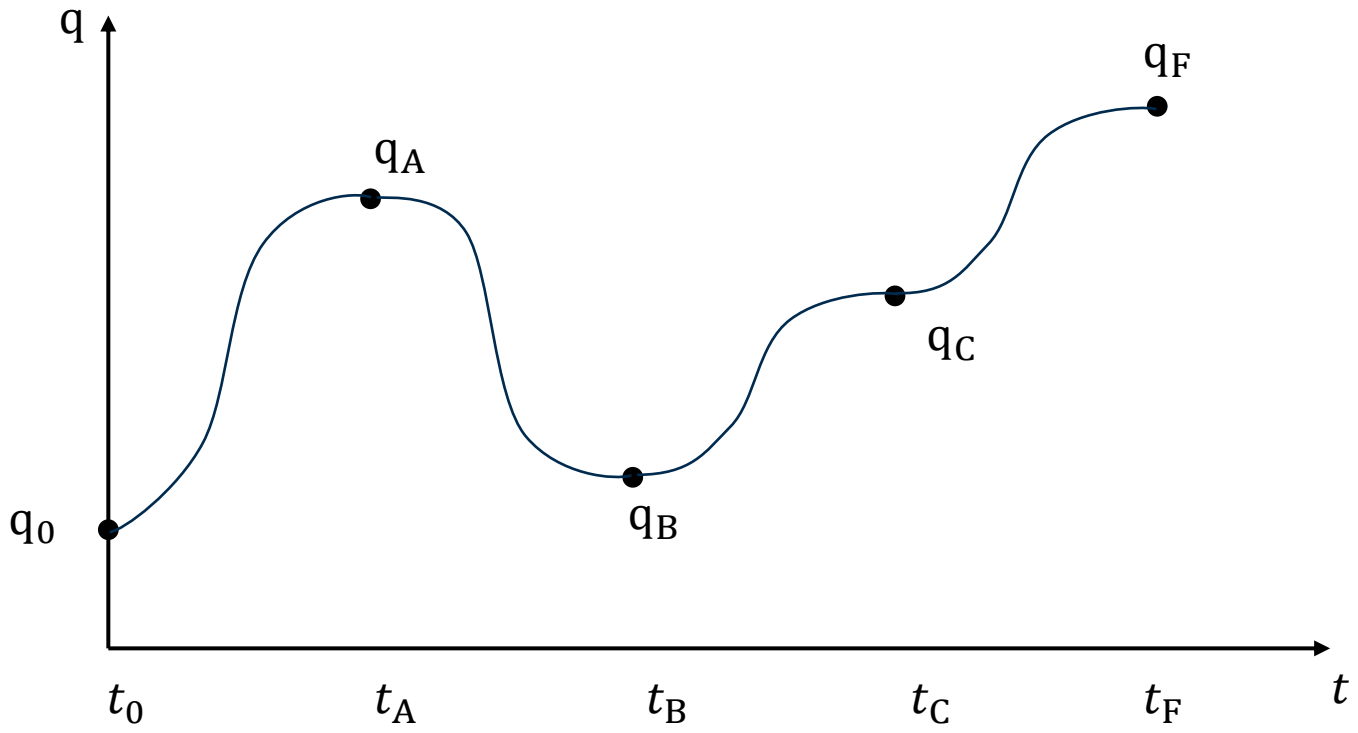
- Sometimes, the user may want to set more points for the robot to pass through on the route to the goal.
- These are called via points.

- How are via points incorporated into the trajectory?



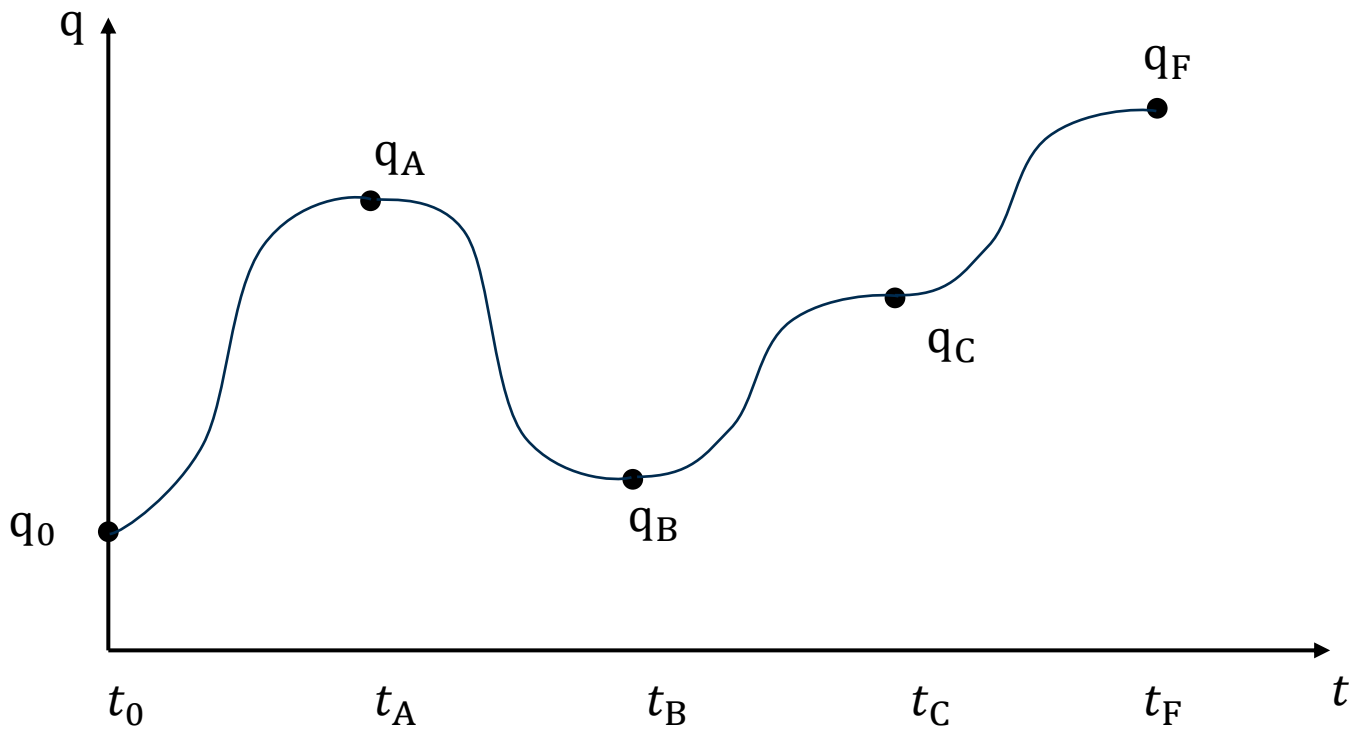
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What are possible ways to include via points?



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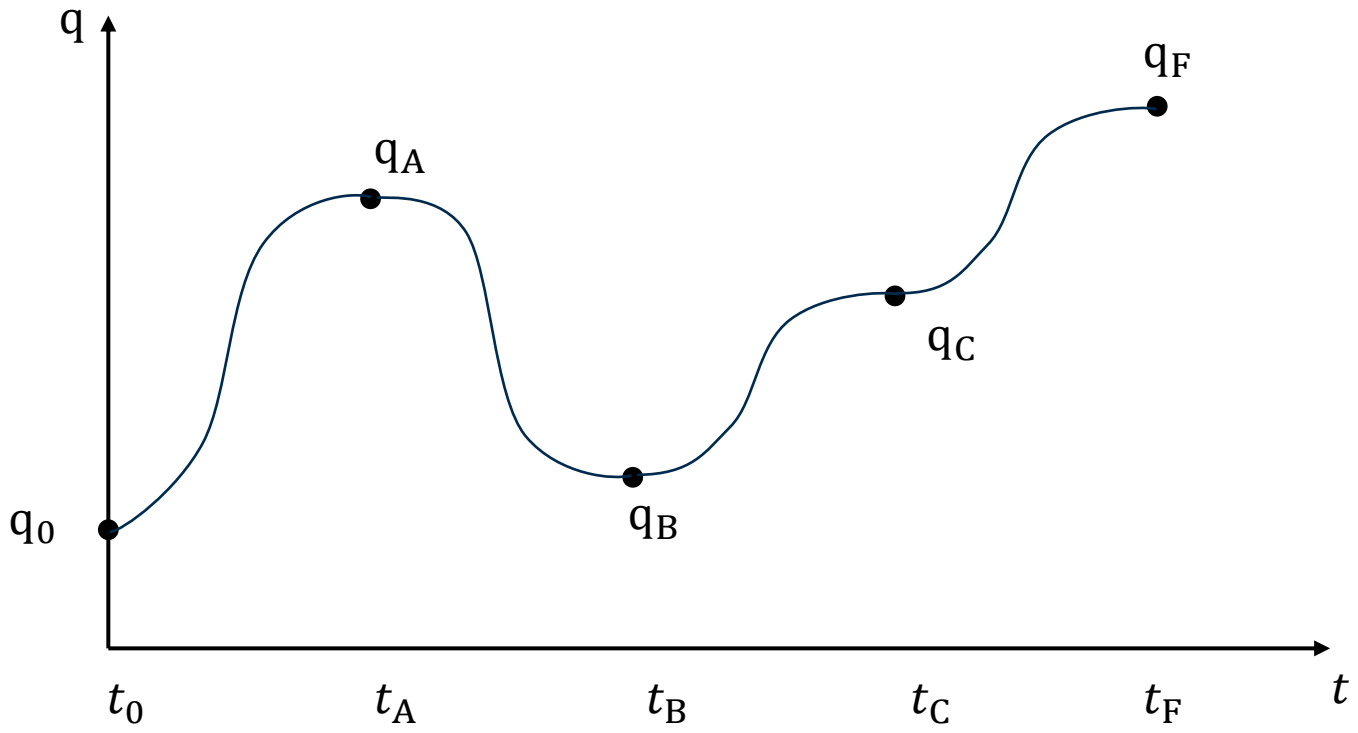
What are possible ways to include via points?



One approach: use a cubic trajectory between each pair of adjacent points

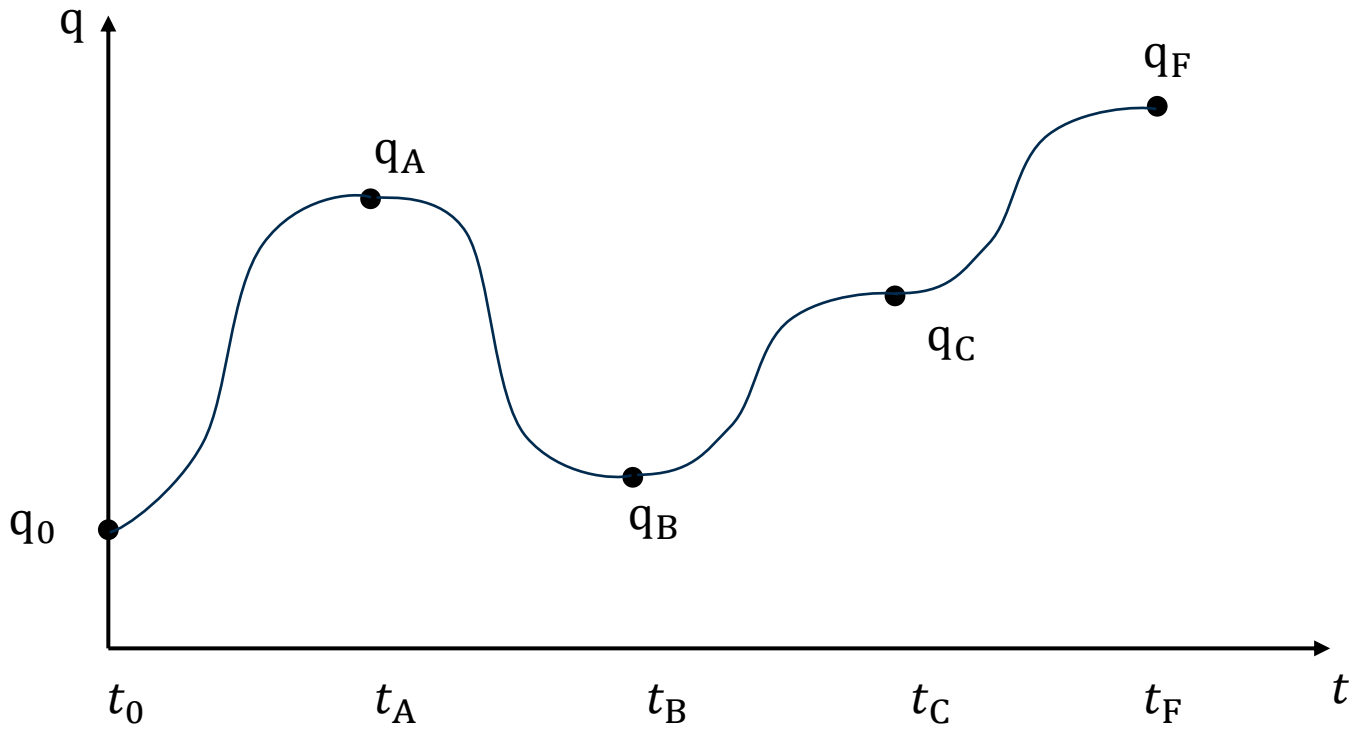
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What about velocities at the via points?



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What about velocities at the via points?



If using a cubic trajectory, like the one we computed earlier, between each consecutive pair of points, the robot comes to rest at each point. Is this desirable?

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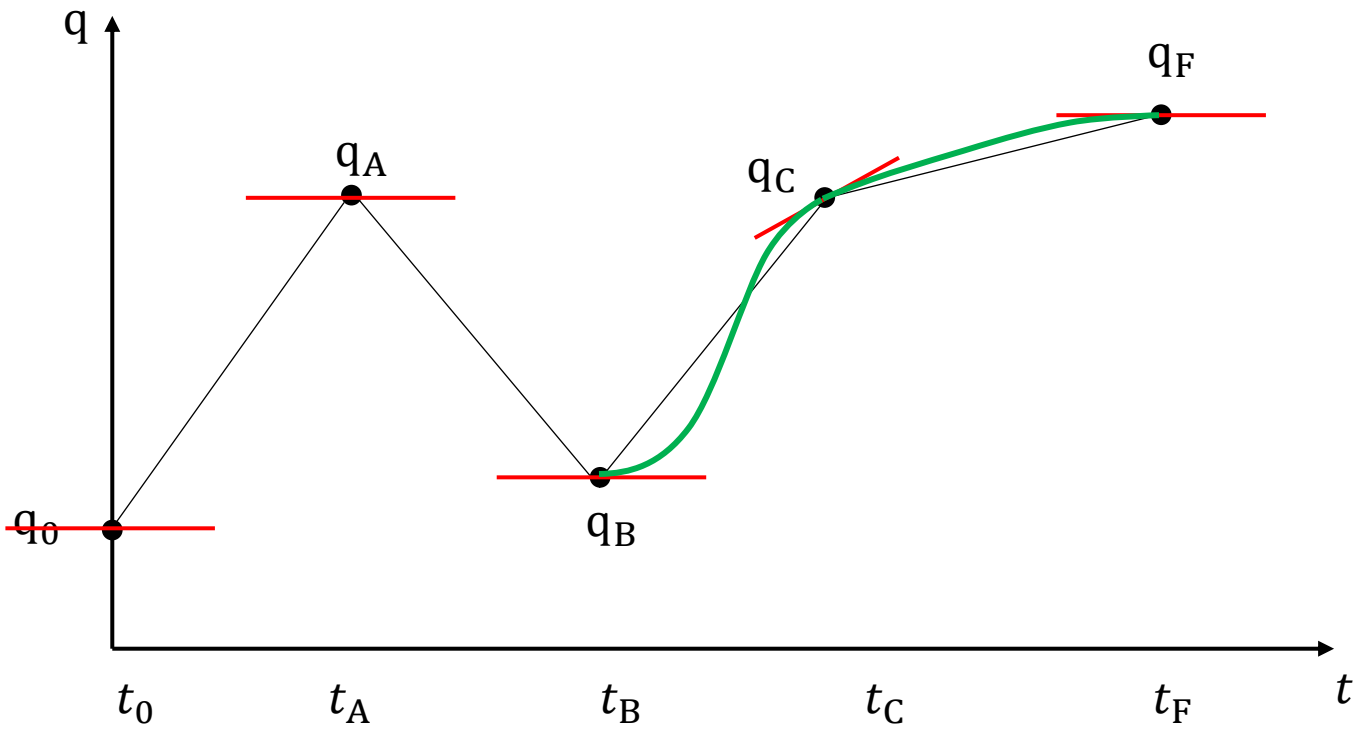
Three ways to deal with velocities

- Option 1: the user explicitly specifies velocities at via points.
- Option 2: the system automatically determines velocities at the via points using a suitable formula (can either be in the joint space or task space).
- Option 3: the system automatically determines velocities at the via points such that the acceleration remains continuous.

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An example of a simple formula

If the slope of the curve changes sign, then assign a zero velocity, else assign the average of the two slopes as the velocity.

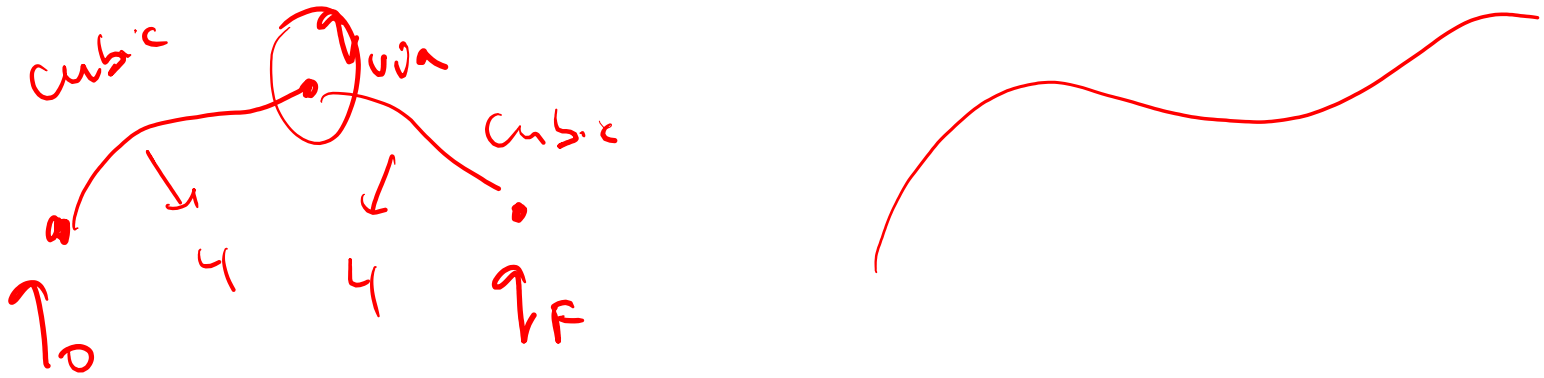


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Homework: constant acceleration

Consider a trajectory with a single via point between the start and the goal. Determine expressions for cubic curves representing the trajectory, if the acceleration is constant at the via point.

Given: q_0, q_{via}, q_F, and $\dot{q}_0 = \dot{q}_F = 0$, and a constant acceleration a_{via}



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Higher order polynomial functions can also be used

Joint position q is described by a quintic polynomial of time *t*

$$\mathrm{q} = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + a_4 t^4 + a_5 t^5$$

We now have 6 unknowns. So, we need 6 equations to solve for them. Where do these come from? These are provided by initial and final conditions on position, velocity, and acceleration.

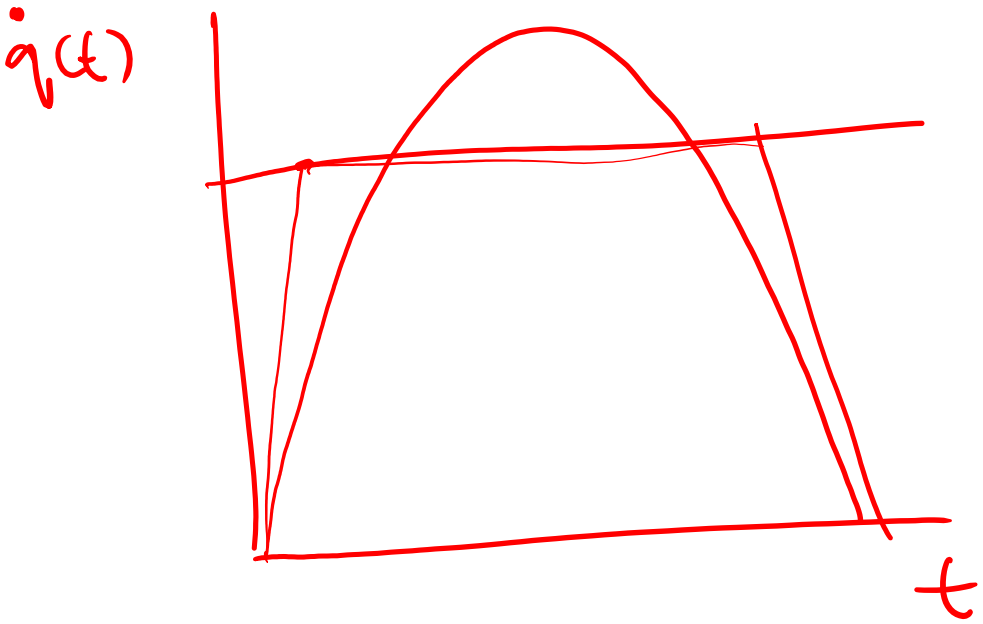
$$\mathrm{q}(t=0) = \mathrm{q}_0 \qquad \mathrm{q}(t=t_F) = \mathrm{q}_F \qquad \dot{\mathrm{q}}(t=0) = \dot{\mathrm{q}}_0 \qquad \dot{\mathrm{q}}(t=t_F) = \dot{\mathrm{q}}_F \qquad \ddot{\mathrm{q}}(t=0) = \ddot{\mathrm{q}}_0 \qquad \ddot{\mathrm{q}}(t=t_F) = \ddot{\mathrm{q}}_F$$

$$\dot{\mathrm{q}} = a_1 + 2a_2 t + 3a_3 t^2 + 4a_4 t^3 + 5a_5 t^4$$

$$\ddot{\mathrm{q}} = 2a_2 + 6a_3 t + 12a_4 t^2 + 20a_5 t^3$$

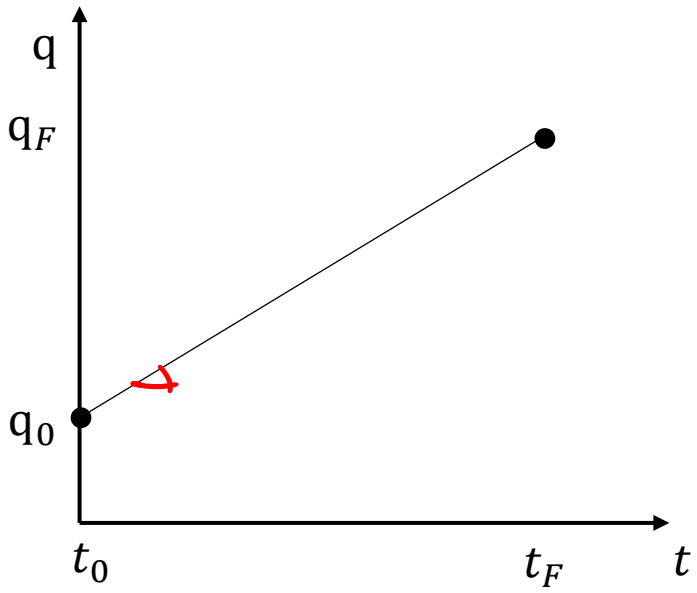
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Polynomial trajectories are inefficient



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Let us revisit linear trajectories

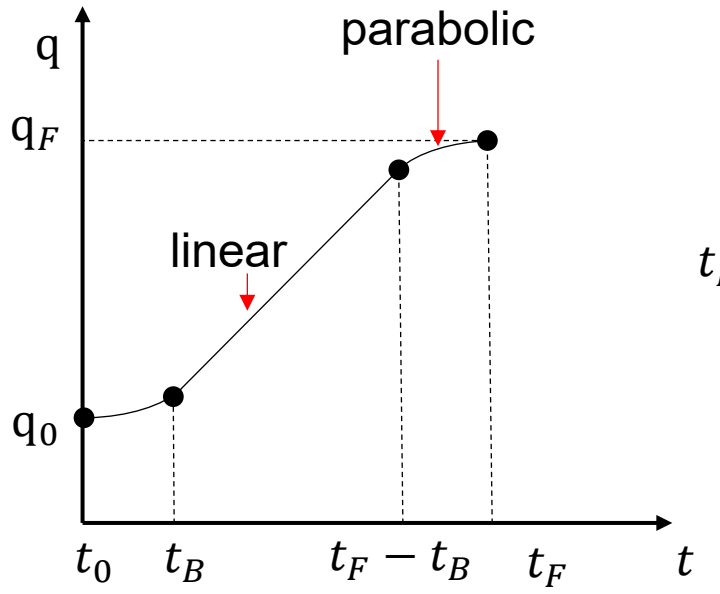


- The simplest linear trajectory is a straight-line interpolation.
- What is the velocity in this case?

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Introduce blending to keep velocities smooth

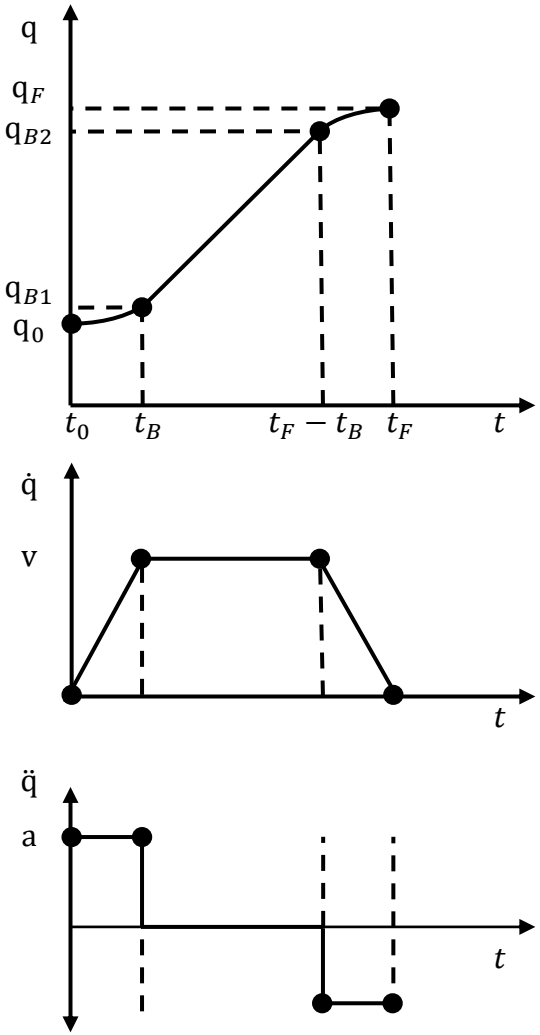
- Example: Linear path with parabolic blends



t_B : duration for which blending is applied

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Linear with parabolic blends



$$q_{B1} = q_0 + \frac{a\,(t_B - t_0)^2}{2}$$ ← Second order polynomial in time

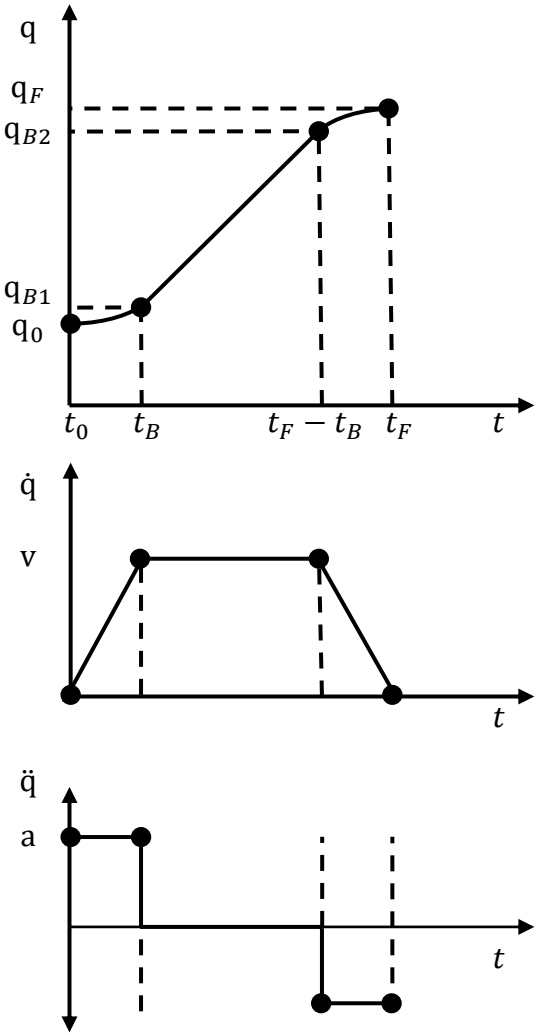
$$q_{B2} = q_{B1} + v(t_F - 2t_B)$$ ← Linear in time

$$q_F = q_{B2} + vt_B - \frac{a\,(t_B)^2}{2}$$ ← Second order polynomial in time

$$t_B = \frac{v}{a}$$

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Linear with parabolic blends



$$q_{B1} = q_0 + \frac{a\,(t_B - t_0)^2}{2}$$

$$q_{B2} = q_{B1} + v(t_F - 2t_B)$$

$$q_F = q_{B2} + v t_B - \frac{a\,(t_B)^2}{2}$$

$$t_B = \frac{v}{a}$$

$$a{t_B}^2 - a(t_F - t_0)t_B + \left(q_F - q_0 - \frac{a\,(t_0)^2}{2}\right) = 0$$

If $t_0 = 0$,

$$a{t_B}^2 - a(t_F)t_B + (q_F - q_0) = 0$$

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What is an appropriate acceleration during blending?

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What is an appropriate acceleration during blending?

$$a\,t_B{}^2 - a(t_F)t_B + (q_F - q_0) = 0$$

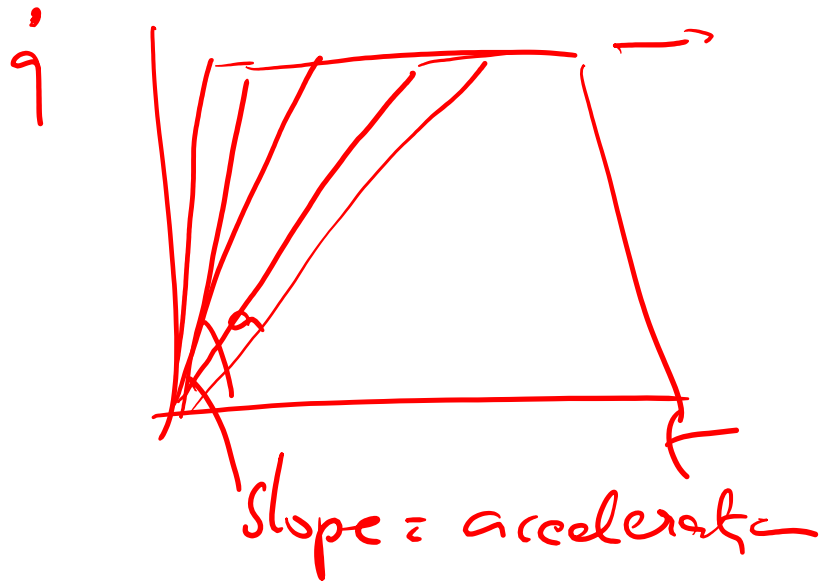
Solve for t_B

$$t_B = \frac{t_F}{2} \pm \frac{\sqrt{a^2(t_F)^2 - 4a(q_F - q_0)}}{2a}$$

For a solution to exist,

$$a^2(t_F)^2 - 4a(q_F - q_0) \geq 0$$

$$a \geq \frac{4(q_F - q_0)}{(t_F)^2}$$



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Example

A joint moves a total for 120 degrees in 2 seconds. Acceleration and deceleration is 300 degrees/s^2. Calculate constant velocity value.

Given: $q_F - q_0 = 120, t_0 = 0, t_F = 2, a = 300 \ ^\circ/s^2$

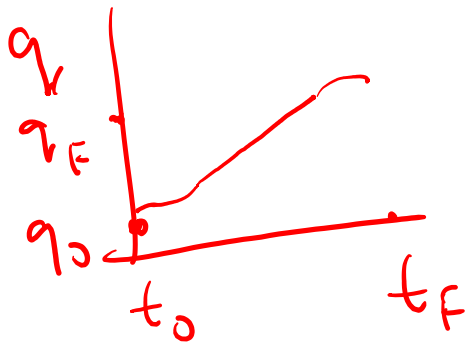
Step 1: verify if the value of a makes sense. Is $a \geq \frac{4(q_F - q_0)}{(t_F)^2}$?

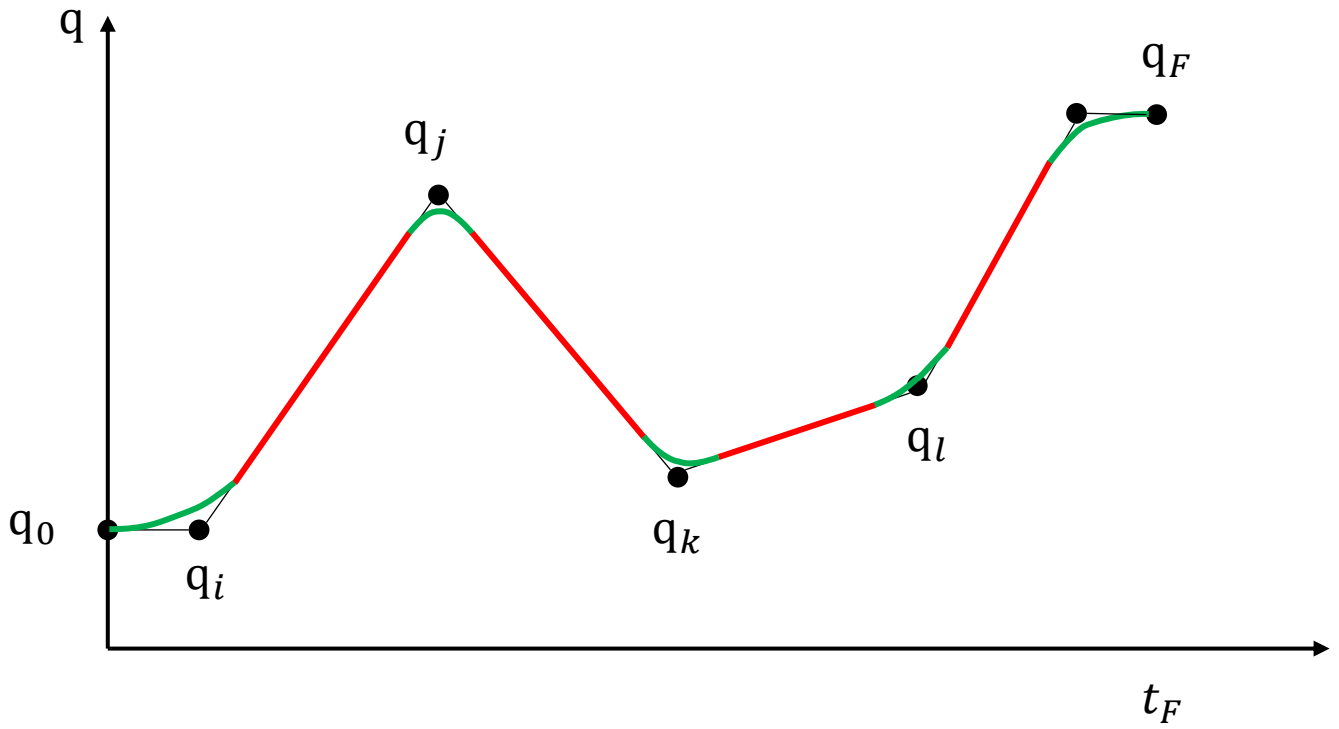
Step 2: If yes, solve for t_B. $t_B = \frac{t_F}{2} \pm \frac{\sqrt{a^2(t_F)^2 - 4a(q_F - q_0)}}{2a}$

Step 3: Solve for v. $v = at_B$

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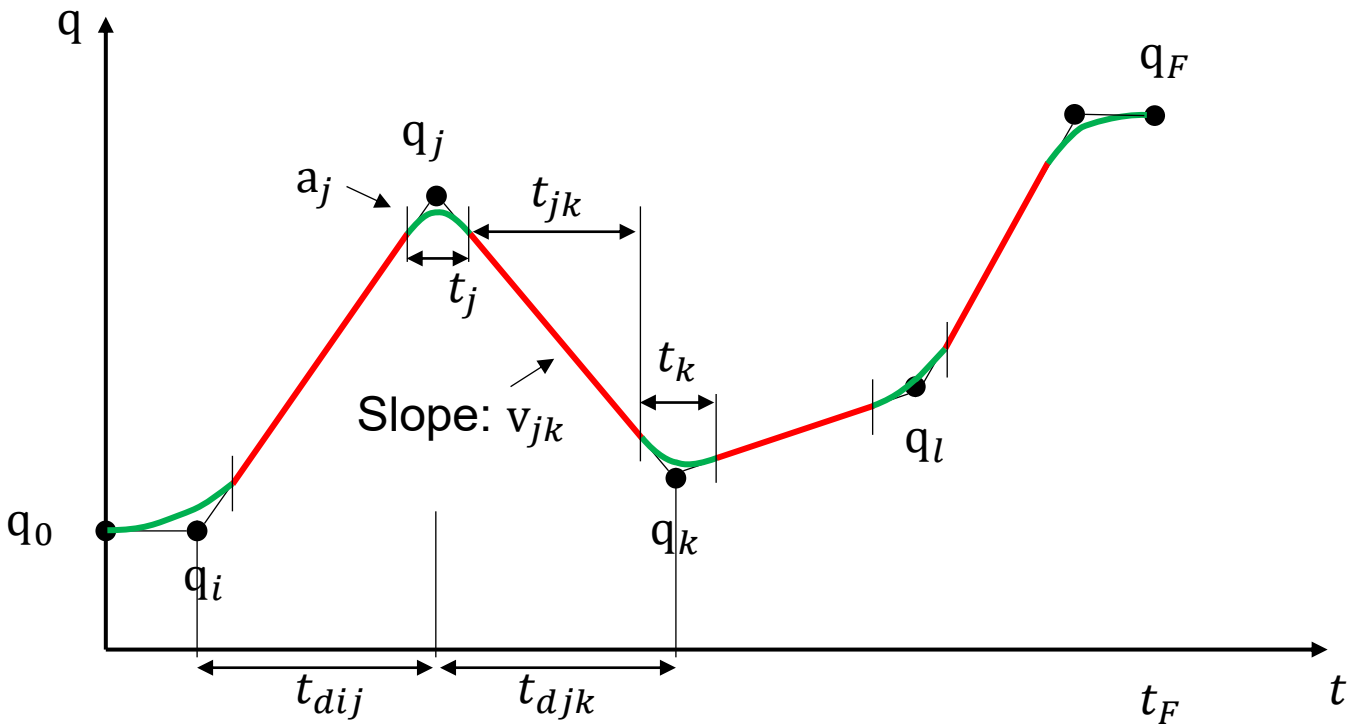
Multi-segment linear paths with blends





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Multi-segment linear paths with parabolic blends



Consider three points at q_i,q_j and q_k.

t_{djk} : the total duration of the path between q_j and q_k

t_{jk} : the duration of the linear path between q_j and q_k

t_j : the duration of the blend at q_j

v_{jk} : the slope of the linear path between q_j and q_k

a_j : the acceleration during blend at q_j

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The setup of the path generation problem

We are given

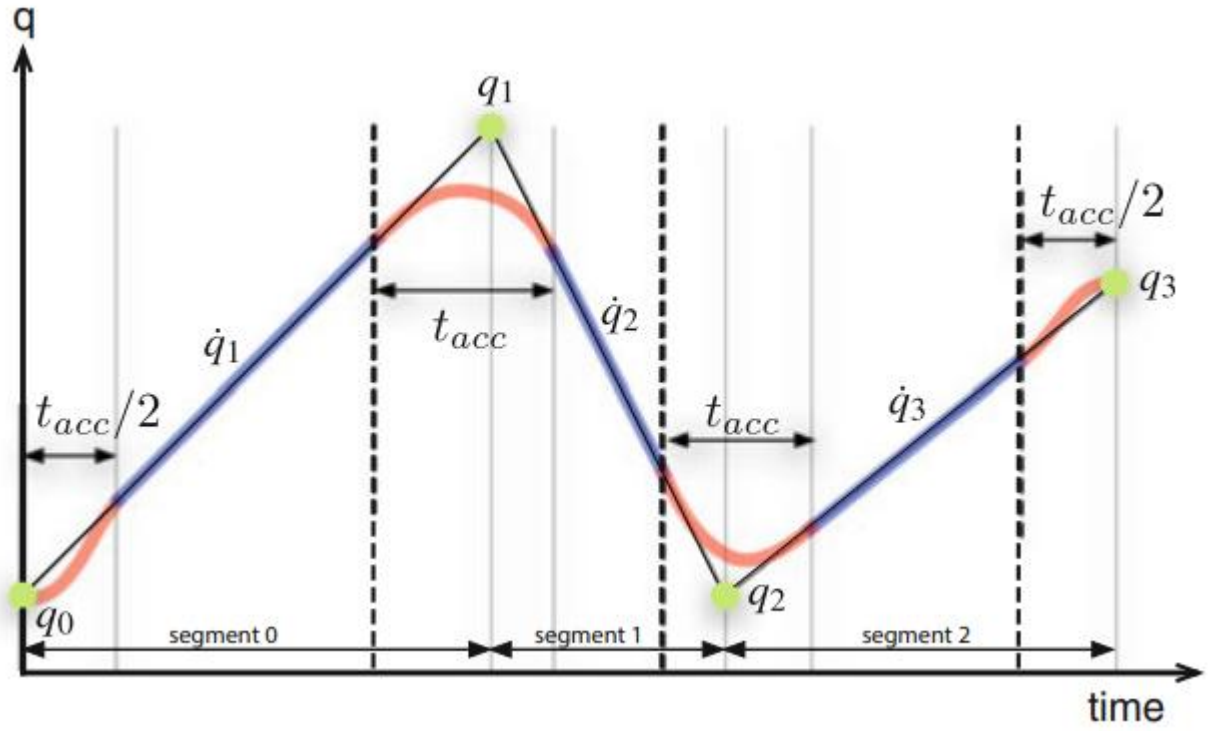
- set of via points, q_k
- the total durations of the individual paths, t_{djk}
- the magnitudes of accelerations to use for parabolic blending at each via point, $|\mathrm{a}_j|$

We would like to determine the blend times t_k

Refer to section 7.3 of Craig's book for a full solution

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Multi-segment linear paths with polynomial blends



Refer to section 3.3.3 of Peter Corke's book

The function mstraj from the robotics toolbox implements this

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Task (Cartesian) Space Trajectories



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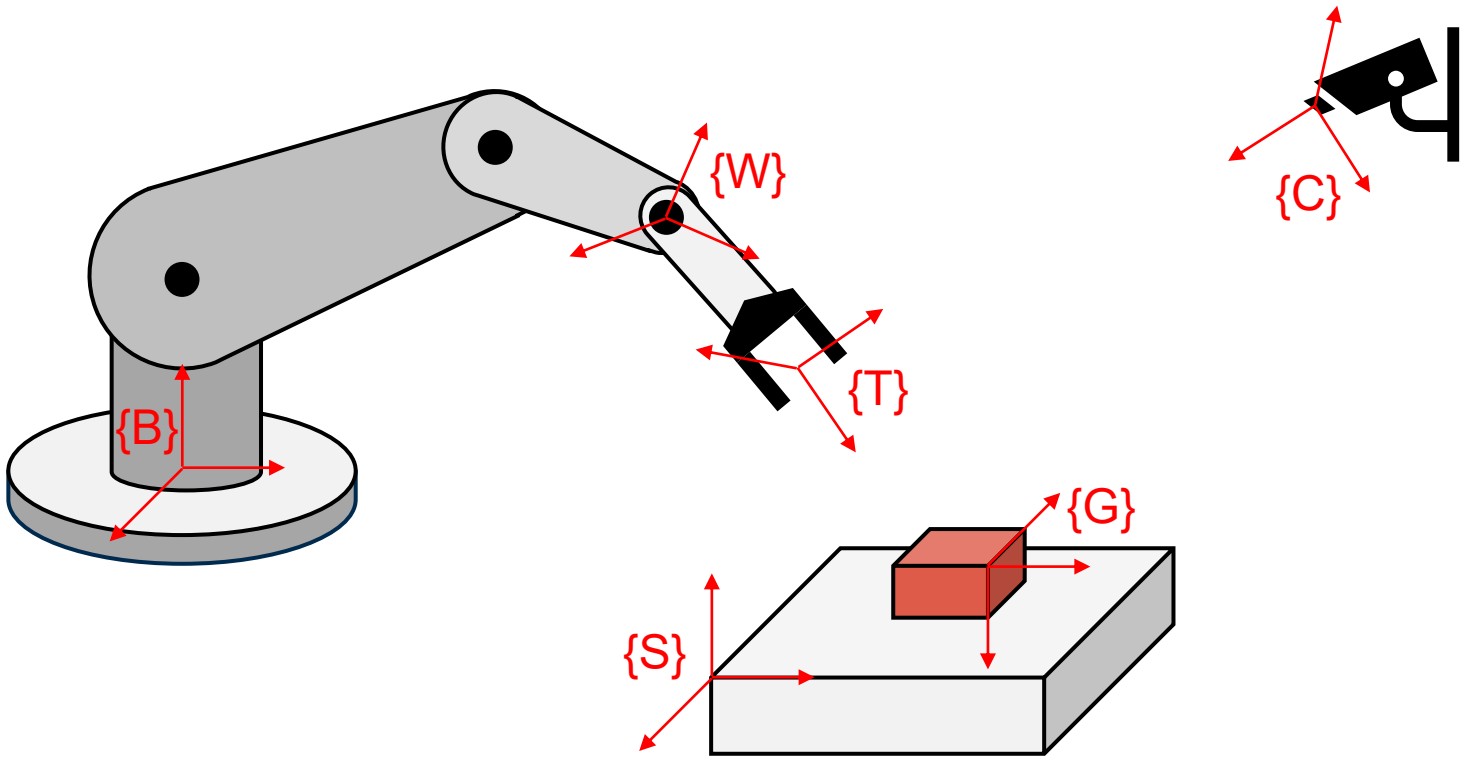
When is task space path generation preferred?

- When a specific path (such as a straight line or an ellipse) is desired for the end effector.
- When avoiding obstacles during motion

Cartesian path generation is computationally more intensive than joint based scheme.

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How does one specify a robot's trajectory?



{B} : Base frame
{W} : Wrist frame
{T} : Tool frame
{S} : Station frame
{G} : Goal frame
{C} : Camera frame

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A common choice for representing trajectories is

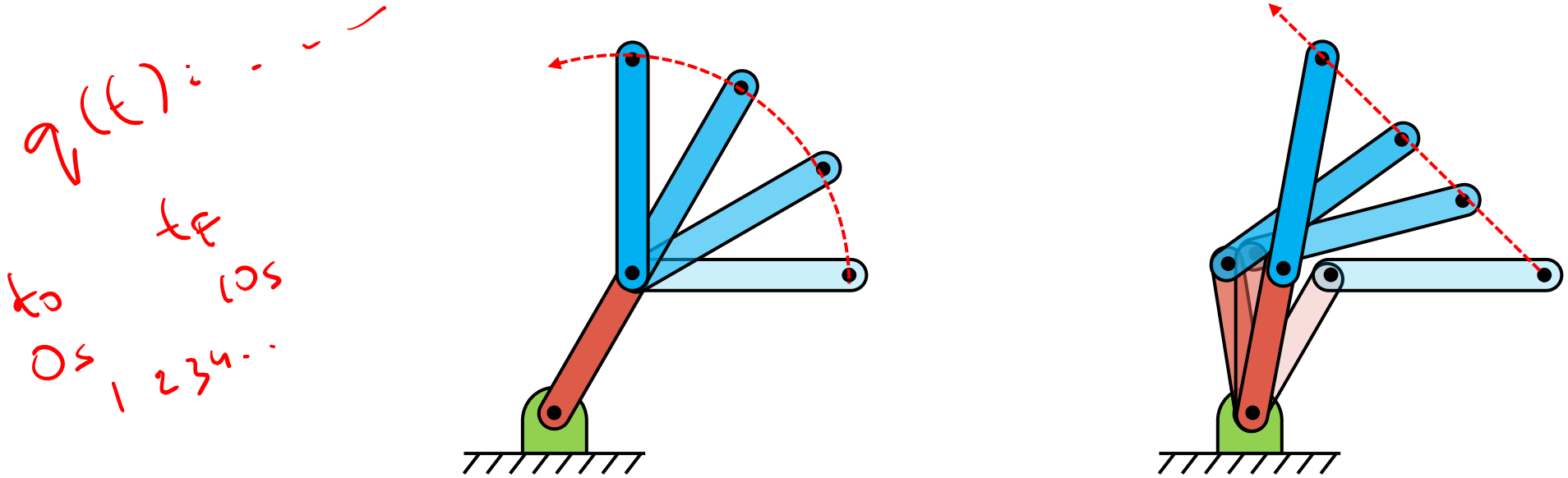
to describe it as the motions of the tool {T} in terms of the station frame {S}.

What are the benefits?:

- The description of the trajectory is decoupled from that of the manipulator – i.e., the same description can be used for various robots or tools.
- The trajectory can be easily adapted for changing {S} – e.g., a conveyor belt.

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Difference in linear interpolation of trajectory between joint space and task space

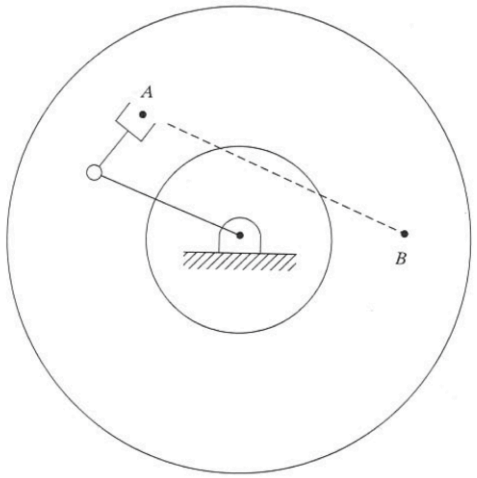


Linear interpolation in joint space

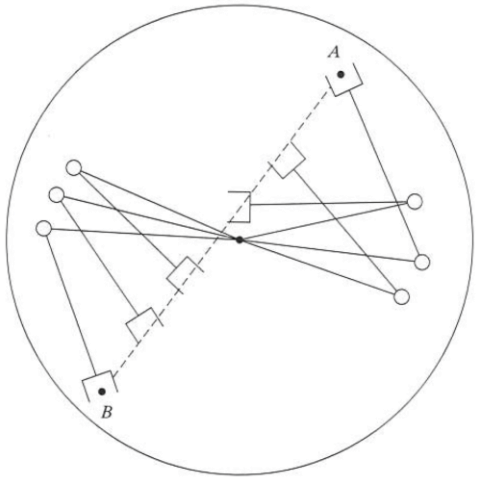
Straight line path in task space

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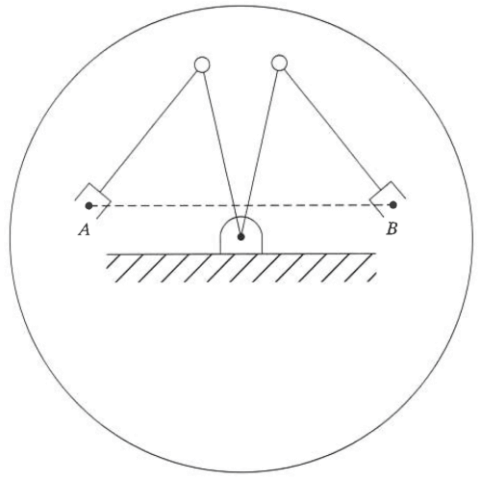
Problems with straight line trajectories in cartesian space



Intermediate points may lie outside the workspace



The path may pass through a singularity



Start and goal are reachable in different configurations

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Task space trajectory generation

Given: a trajectory for the tool in cartesian coordinates

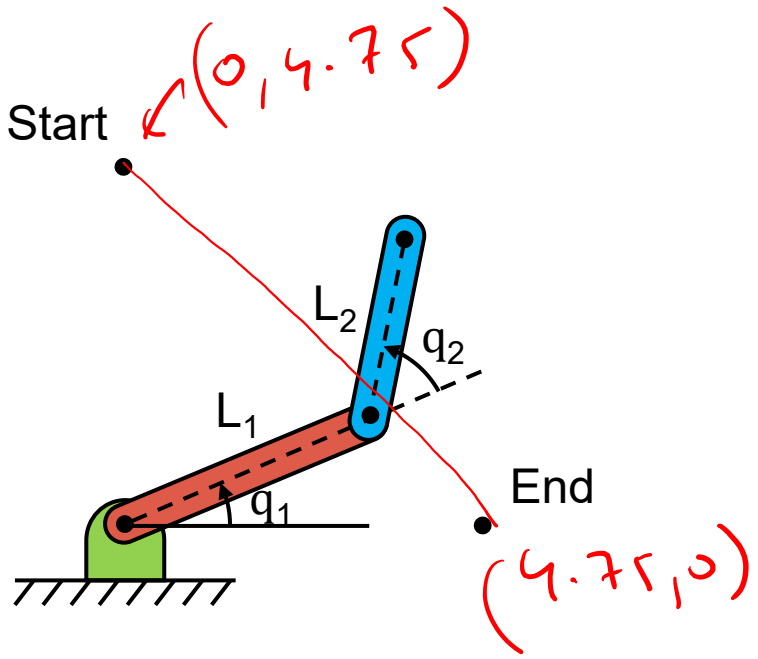
Aim: obtain a series of joint positions to send to the motors

Method:

- Use inverse kinematics of the manipulator for each given cartesian pose of the tool to obtain the corresponding joint positions.
- Fit a curve over the joint positions.
- Resample from the curve to obtain finer joint positions.

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Example: planar 2 link RR manipulator



Given:

- $L_1 = 4$ m, $L_2 = 3$ m
- Start point: $(0, L_1 + 0.25\, L_2) = (0, 4.75)$
- End point: $(L_1 + 0.25\, L_2, 0) = (4.75, 0)$
- The manipulator must move at a constant speed of 3 m/s
- The tool must follow a straight-line path

Goal:

- Generate a set of joint coordinates that accomplish this movement

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Step 1: Formulate a function for the path

$y = -1x + 4.75$

The path is a straight-line between $(0,4.75)$ and $(4.75,0)$.

What is the equation of the path?

$$\frac{y - 4.75}{0 - 4.75} = \frac{x - 0}{4.75 - 0}$$

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Step 1: Formulate a function for the path

The path is a straight-line between $(0,4.75)$ and $(4.75,0)$.

Equation of the line that satisfies this path is: $y = -x + 4.75$

Length of the path is 6.7175 m.

If moving at a speed of 3 m/s, time taken to traverse the path is 2.24 s.

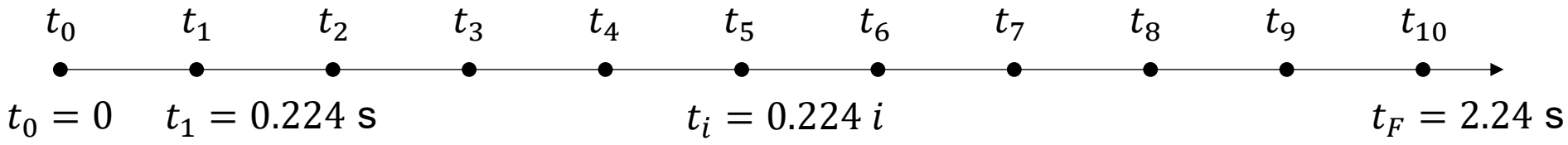
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Step 2: Sample the function for intermediate points

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Step 2: Sample the function for intermediate points

Say we split the total travel time into 10 equal sections



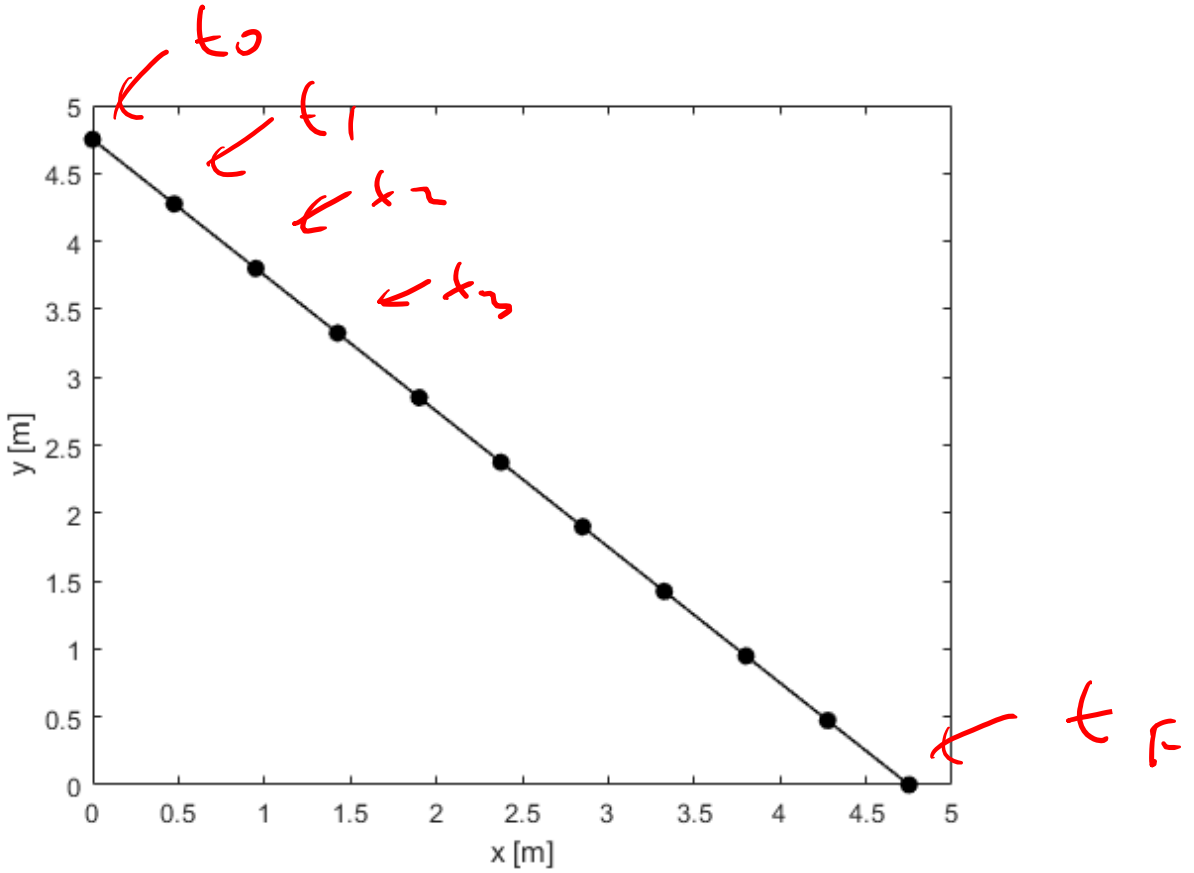
For motion at constant velocity v along a generic line $\mathrm{y} = \mathrm{mx} + \mathrm{c}$, we can write

$$\mathrm{x}(t) = \mathrm{x}_0 + \frac{\mathrm{v}t}{\sqrt{1+\mathrm{m}^2}} \qquad \mathrm{y}(t) = \mathrm{y}_0 + \frac{\mathrm{vm}t}{\sqrt{1+\mathrm{m}^2}}$$

$$\mathrm{x}(t_i) = \frac{3t_i}{\sqrt{2}} \qquad \mathrm{y}(t_i) = 4.75 - \frac{3t_i}{\sqrt{2}}$$

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The sampled points are as follows



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Step 3: Use IK to obtain joint positions for each sampled cartesian position

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Step 3: Use IK to obtain joint positions for each sampled cartesian position

Recall IK for positions that we derived in week 3:

$$q_2 = \text{atan2}(\sin(q_2), \cos(q_2))$$

where,

$$\cos(q_2) = \frac{x^2 + y^2 - L_1^2 - L_2^2}{2L_1L_2}$$

$$\sin(q_2) = \pm\sqrt{1 - \cos(q_2)^2}$$

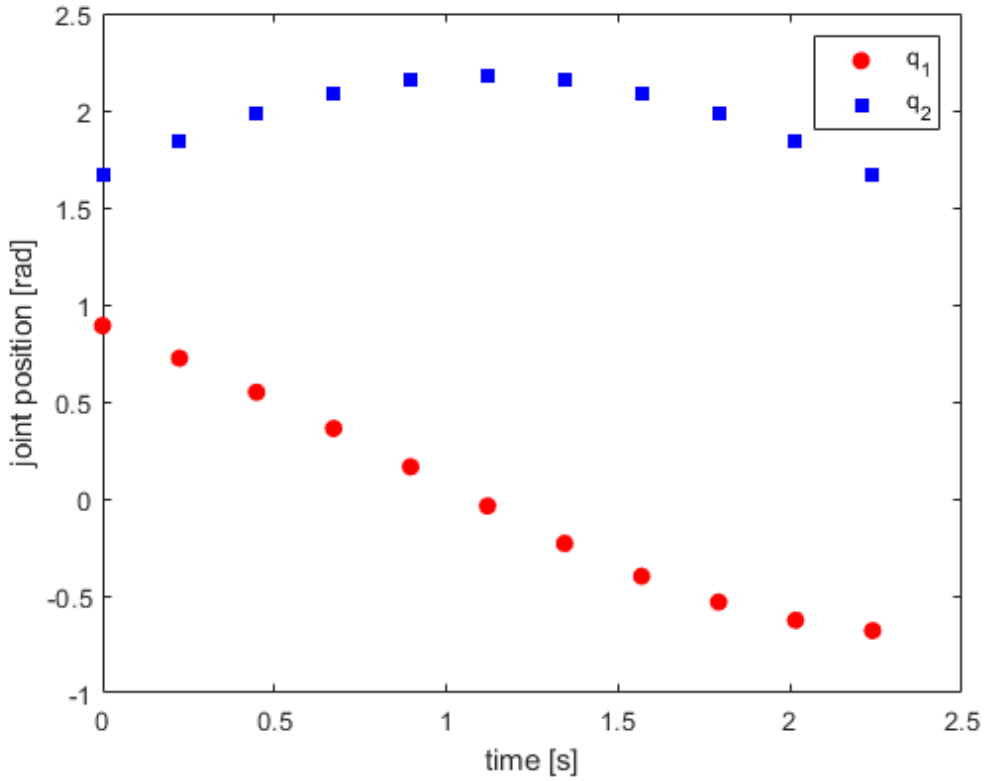
and

$$q_1 = \text{atan2}(y, x) - \text{atan2}(L_2 \sin(q_2), L_1 + L_2 \cos(q_2))$$

Note: If velocity is not constant, obtain it at each point using the Jacobian

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The joint positions are as follows

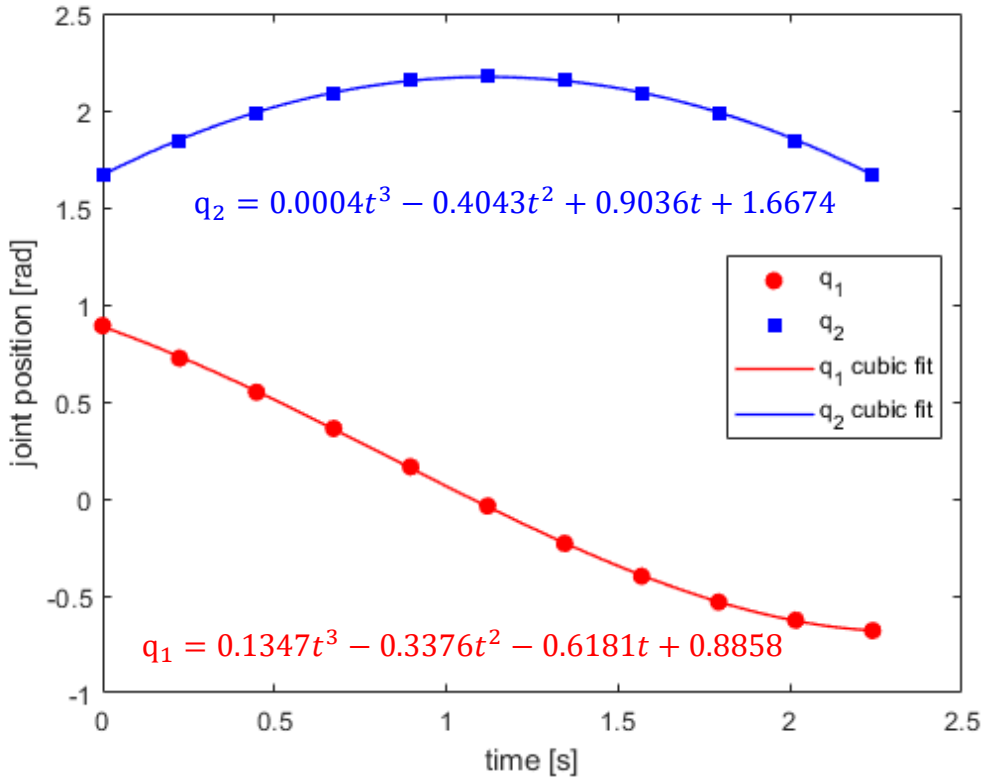


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Step 4: Fit a curve to the joint positions

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Step 4: Fit a curve to the joint positions



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A note about fitting polynomials

- A single cubic may not be sufficient for the entire trajectory.
- A quintic can be used to fit curves for position, velocity and acceleration.

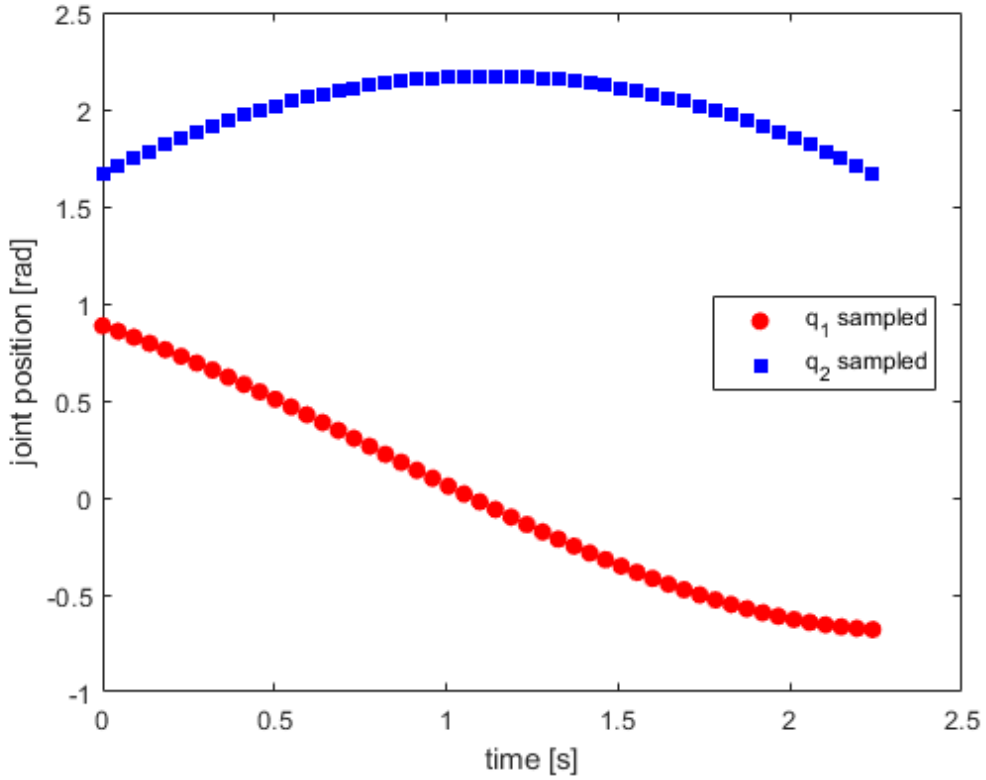
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Step 5: Sample from the fit curves

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Step 5: Sample from the fit curves

Each curve was sampled for 50 points

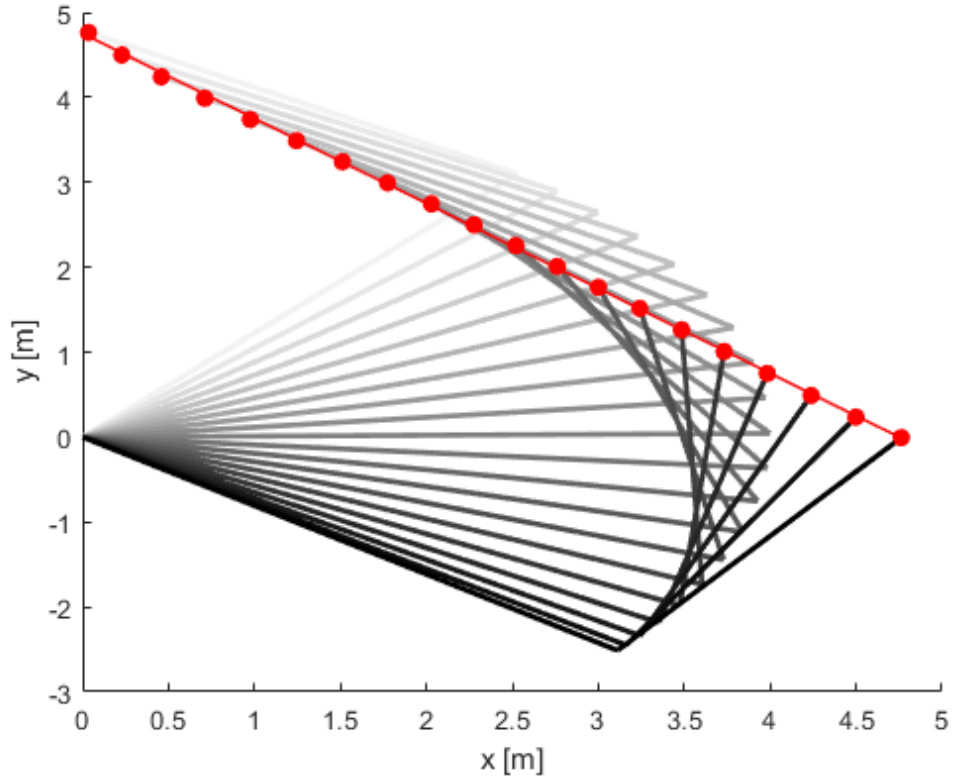


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Step 6: Use FK to get the trajectory

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Step 6: Use FK to get the trajectory



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What if you want to interpolate between poses?

- You can directly interpolate between positions and angles
- You cannot use simple interpolation between translation matrices or rotation matrices.

- Refer to chapter 3 of Peter's book.
- The ctraj function from the toolbox is useful – see the Jupyter notebook.

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What can you do, now that you know how to generate trajectories?

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Next assessment is on trajectories

- You will be given a description of the path – combination of straight lines and curves in 3D space.

- Generate trajectories to make the robot's tool follow the path

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Summary

- Initial position and goal are usually specified in cartesian space
  - Pose of {T} with respect to {S}
- The path in between can be generated in either joint space or task (cartesian) space.
- Whichever scheme is used, commands to the motor are always provided in terms of joint positions, velocities and accelerations.

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Next week

- Velocities and Jacobians

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